# SQ9 Auto Plan Compiler v0.2

## Mathematical Audit and US Equities Price-Mapping Resolution

Assessment only - no application implementation

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Assessment date: 11 October 2026

Scope: Mathematical assessment only; no application implementation

Authorial role: Quantitative mathematics, financial systems, and Python algorithm audit

## EXECUTIVE CONCLUSION

The supplied Python v0.2 calculation engine was analyzed directly and reproduced in temporary standalone Python executions. The audit did not substitute a conventional Square of Nine formula for the supplied implementation.

Principal findings:

1. The 19 x 19 spiral is internally correct. It contains exactly 361 unique cells and exactly the ordinals 1 through 361, with the documented orientation.
2. There are exactly 73 harmonic ordinals: the center plus 72 points distributed evenly across eight rays.
3. Under the validated BUY and SELL pivot relationships, BUY and SELL have identical normalized ordinal feasibility. Both always use adverse ordinal direction -1.
4. A complete stop pair exists precisely when EP1 is one of the 64 harmonics from ordinal 11 through 361. In an accepted plan with a target still available, EP1 is necessarily at most 352.
5. Define D = abs(Close - Pivot), a = abs(Increment), u = D/a, and v = Norm/a. A legacy-accepted plan exists exactly when 10 < u < 360. A full six-level plan, including TP2, exists exactly when 10 < u < 351 and u + v <= 355.5.
6. The corresponding legacy feasible scale interval is D/360 < a < D/10. The full-plan scale domain is a > D/351, a >= (D+N)/355.5, and a < D/10.
7. Pivot, Close, Norm, and Direction do not uniquely identify the scale. A further calibration convention is mathematically unavoidable.
8. If no new user input may be added, the recommended deterministic convention is a = abs(Close-Pivot)/11, with Increment +a for BUY and -a for SELL. This places Close at normalized coordinate x=12 and makes EP1 ordinal 11, the first harmonic possessing both legacy angular stops. This is an explicit engineering calibration, not a uniquely discovered SQ9 constant and not evidence of trading edge.

9. If an additional manual input is permitted, the more transparent design is to let the user supply the calibrated positive scale a, validate it against the exact feasibility interval, and record it in the result.

# SECTION A - SOURCE CODE AUDIT

## A1. Scope and method

The supplied build_spiral, coordinate map, harmonic selection, price mapping, entry, target, ray, angular stop, and complete plan functions were transcribed without behavioral changes into temporary inline Python programs. The GUI is irrelevant to these calculations.

## A2. Spiral construction

Executed verification returned:

- Coordinate count: 361.
- Unique coordinate count: 361.
- Assigned ordinal count: 361.
- Unique ordinal count: 361.
- Ordinal range: 1 through 361.
- Missing ordinals: none.
- Duplicate coordinates: none.
- Center: ordinal 1 at relative coordinate (0,0).
- Absolute center array position: (9,9).

The first ring is:

```
Coordinate | Ordinal
(-1,-1) | 3
(-1,0)  | 4
(-1,1)  | 5
(0,-1)  | 2
(0,0)   | 1
(0,1)   | 6
(1,-1)  | 9
(1,0)   | 8
(1,1)   | 7
```

Therefore the documented orientation is verified:

3 4 5

2 1 6

9 8 7

Construction classification:

- Verified behavior: correct 361-cell spiral for configured size 19.
- Design assumption: the size is odd and has a unique central cell.
- No v0.2 anomaly: there are no missing or duplicated coordinates or ordinals.
- Generalization caution: correctness for arbitrary even sizes is not established, but is irrelevant to the fixed canonical grid.

## A3. Harmonic classification behavior

An ordinal is harmonic exactly when its relative coordinate satisfies row=0, column=0, or abs(row)=abs(column). This is the union of two cardinal axes and two diagonals. Center ordinal 1 belongs to both cardinal and diagonal predicates but occurs once in HARMONIC_N.

## A4. Price mapping

The exact mapping is:

P(n) = Pivot + (n-1) * Increment.

For original v0.2 behavior:

- BUY requires Pivot < Close and therefore receives Increment +1.
- SELL requires Pivot > Close and therefore receives Increment -1.

Increasing ordinal consequently moves from Pivot toward Close and then toward the profit side in both directions. The increment direction is consistent with pivot validation.

Embedded assumptions:

1. One ordinal step is exactly one price unit.
2. The same scale applies throughout the grid.
3. The price embedding is affine, with no nonlinear transformation.
4. Pivot is assigned to ordinal 1.
5. Increment sign is fixed by Pivot-to-Close orientation.
6. Increment magnitude 1 is hard-coded and is not derived from grid geometry.
7. Harmonic membership is ordinal-based, not quotation-tick-based.
8. Negative generated prices are not rejected.
9. Prices use binary floating point.

## A5. Entry algorithm

Let ordered harmonics be h1 < h2 < ... < h73, with h1=1 and h73=361. For generalized positive scale a define x=1+D/a=1+u.

For both valid BUY and valid SELL, an entry harmonic is eligible exactly when h_i < x. This is a strict inequality. Eligible harmonics are ordered nearest to Close first. EP1 is the largest harmonic below x and EP2 is the second largest.

Exact piecewise rule: EP1=h_j and EP2=h_(j-1) exactly when h_j < x <= h_(j+1), equivalently h_j-1 < u <= h_(j+1)-1. At x equal to a harmonic h_k, h_k is excluded; the pair remains h_(k-1), h_(k-2).

For valid pivots u>0. No eligible entry is then impossible. Exactly one entry exists when 0<u<=1. At least two entries exist exactly when u>1.

## A6. Target algorithm

RawTP is Close+Norm for BUY and Close-Norm for SELL. In normalized coordinates both become y=x+v=1+u+v.

Profit-side harmonics satisfy h_i>x. TP1 minimizes abs(h_i-y). The secondary tuple key minimizes abs(h_i-x). Because all eligible candidates are above x in normalized ordinal space, an exact midpoint tie selects the lower ordinal, which is closer to Close.

TP2 is the immediately following eligible harmonic beyond TP1. TP1 exists exactly when x<361, equivalently u<360. RawTP may lie beyond grid coverage without causing TP1 failure: the nearest eligible harmonic, often 361, is still returned. If TP1 is 361, TP2 is None, and calculate_plan accepts the plan.

## A7. Stop-loss algorithm

ray_of_n returns None for center ordinal 1 and non-harmonics. Otherwise it reduces row and column to signs, identifying one of eight discrete rays. n_on_ray returns the harmonic ordinals on a ray.

adverse_ordinal_sign gives:

```
Direction | Increment sign | Adverse ordinal sign
BUY       | positive       | -1
BUY       | negative       | +1
SELL      | positive       | +1
SELL      | negative       | -1
```

All inputs admitted by calculate_plan use -1: BUY has positive Increment and SELL has negative Increment.

For actual validated cases, the 181 stop forms theoretical target (sqrt(e)-1)^2, searches the opposite ray, requires candidate n<e, and selects the nearest available candidate. The 361 stop forms theoretical target (sqrt(e)-2)^2, searches the same ray, requires n<e, and selects the nearest candidate.

The returned ordinal is not assumed to equal the theoretical square-root target. The theoretical value is only the center of a nearest-candidate search constrained by harmonic membership, ray, adverse ordinal direction, and the finite grid. The 181 and 361 labels describe these discrete implementation rules, not literal continuous rotations of a stock price.

Implementation defect: root_delta = 1 if degrees == 181 else 2. Every unsupported degree silently receives root displacement 2, and every value other than exactly 181 uses the same-ray branch. This is a validation defect, not permission to alter the intended two stop methods.

### A8. Classification verdict

```
Finding | Classification
Correct 361-cell size-19 spiral | Verified behavior
Correct documented orientation | Verified behavior
73 union harmonics | Verified behavior
Pivot fixed at ordinal 1 | Design assumption
One-dollar increment magnitude | Design assumption
Affine price mapping | Design assumption
Strict entry and target comparisons | Verified behavior
TP2 may be None while plan succeeds | Verified behavior
Nearest ray candidate may differ from theoretical square | Mathematical limitation
Finite ray/grid boundaries can prevent stops | Mathematical limitation
Unsupported degrees use the non-181 branch | Implementation defect
No positivity check on generated levels | Implementation defect / omitted domain validation
Float comparisons and outputs | Implementation limitation
Selecting an equity scale | Requires external methodological decision
Predictive value or profitability | Requires external empirical decision
```

## SECTION B - COMPLETE HARMONIC GEOMETRY

### B1. Exact harmonic ordinals

There are 73 harmonics:

1,

2, 3, 4, 5, 6, 7, 8, 9,

11, 13, 15, 17, 19, 21, 23, 25,

28, 31, 34, 37, 40, 43, 46, 49,

53, 57, 61, 65, 69, 73, 77, 81,

86, 91, 96, 101, 106, 111, 116, 121,

127, 133, 139, 145, 151, 157, 163, 169,

176, 183, 190, 197, 204, 211, 218, 225,

233, 241, 249, 257, 265, 273, 281, 289,

298, 307, 316, 325, 334, 343, 352, 361.

Minimum harmonic is 1. Maximum is 361. Cardinal membership including center is 37. Diagonal membership including center is 37. Their intersection is center 1 only, so the union count is 37+37-1=73.
Excluding center there are 36 cardinal and 36 diagonal harmonics.

### B2. Ray distribution

```
Ray | Description | Count
(0,-1) | west      | 9
```

```
(-1,-1)| northwest | 9
(-1,0) | north     | 9
(-1,1) | northeast | 9
(0,1)  | east      | 9
(1,1)  | southeast | 9
(1,0)  | south     | 9
(1,-1) | southwest | 9
None    | center    | 1
```

The outside-center distribution is exactly uniform.

## B3. Exact coordinate representation for every harmonic

For every ring k=1,...,9, the eight harmonics are:

```
Coordinate | Type | Ordinal
(0,-k)   | cardinal west      | 4k^2-3k+1
(-k,-k)  | diagonal northwest | 4k^2-2k+1
(-k,0)   | cardinal north     | 4k^2-1
(-k,k)   | diagonal northeast | 4k^2+1
(0,k)    | cardinal east      | 4k^2+2k-1
(k,k)    | diagonal southeast | 4k^2+2k+1
(k,0)    | cardinal south     | 4k^2+4k-1
(k,-k)   | diagonal southwest | (2k+1)^2
```

Together with center (0,0)->1, these formulas enumerate the coordinate, type, ray, and ordinal of every harmonic.

## B4. Spacing

The consecutive harmonic gaps are eight gaps of 1, eight gaps of 2, and so on through eight gaps of 9. Harmonic spacing is therefore nonuniform in ordinal space.

## B5. Complete angular mappings: adverse direction -1

This is the direction used by every valid BUY and SELL plan.

- Center 1: neither stop.
- Ring-one ordinals 2,3,4,5: neither stop.
- Ring-one ordinals 6,7,8,9: SL181 only.
- Every harmonic on rings 2 through 9, ordinals 11 through 361: both stops.

```
EP1 | Coordinate | Ray | SL181 | SL361 | Support
1 | (0,0) | none | - | - | neither
2 | (0,-1) | west | - | - | neither
3 | (-1,-1) | northwest | - | - | neither
4 | (-1,0) | north | - | - | neither
5 | (-1,1) | northeast | - | - | neither
6 | (0,1) | east | 2 | - | 181 only
7 | (1,1) | southeast | 3 | - | 181 only
8 | (1,0) | south | 4 | - | 181 only
9 | (1,-1) | southwest | 5 | - | 181 only
```

For every ring k>=2, SL181 is the harmonic on the opposite ray of ring k-1, while SL361 is the harmonic on the same ray of ring k-1. This compact rule determines both candidates for every remaining harmonic.

Examples:

```
EP1 | Coordinate | SL181 | SL361
11  | (0,-2)  | 6   | 2
13  | (-2,-2) | 7   | 3
15  | (-2,0)  | 8   | 4
17  | (-2,2)  | 9   | 5
19  | (0,2)   | 11  | 6
25  | (2,-2)  | 17  | 9
343 | (9,9)   | 307 | 273
352 | (9,0)   | 316 | 281
361 | (9,-9)  | 325 | 289
```

Exact support counts for adverse -1:

Both: 64

SL181 only: 4

SL361 only: 0

Neither: 5

### B6. Complete angular mappings: adverse direction +1

This direction cannot arise from the validated pivot/direction relationship, but was exhaustively checked.

- Rings 1 through 8: both stops, 64 harmonics.
- Ring-9 west through northeast, ordinals 298,307,316,325: SL181 only.
- Center and ring-9 east through southwest, ordinals 1,334,343,352,361: neither.
- There are no SL361-only cases.

Counts are again 64 both, 4 SL181-only, 0 SL361-only, and 5 neither. The count symmetry hides a boundary reversal: unusable ordinals occur at the low end for adverse -1 and at the high end for adverse +1.

## SECTION C - EP/TP/SL FEASIBILITY

### C1. BUY/SELL symmetry

BUY has Pivot<Close and Increment +a. SELL has Pivot>Close and Increment -a. Both yield x=1+D/a. Both have entries n<x, targets n>x, RawTP coordinate y=x+v, and adverse ordinal sign -1. Therefore BUY and SELL feasibility is identical in normalized ordinal space. Dollar prices are reflections about the relevant Pivot.

### C2. Entry and stop intervals

```
Normalized u | Entry result | Stop result
0<u<=1 | EP1 only | rejected before stops
1<u<=5 | EP1/EP2; EP1 in 2..5 | neither stop
5<u<=10 | EP1/EP2; EP1 in 6..9 | SL181 only
u>10 | EP1/EP2; EP1 at least 11 | both stops, subject to target boundary
```

At u=10, x=11 and strict entry comparison excludes harmonic 11. EP1 remains 9, so SL361 remains unavailable. The condition is u>10, never u>=10.

### C3. Target feasibility

At least one target and TP1 exist exactly when x<361, equivalently u<360.

TP2 exists unless TP1 is 361. For TP1 not to be 361, ordinal 352 must be eligible, requiring u<351, and RawTP coordinate must not exceed the midpoint between 352 and 361:

y <= (352+361)/2 = 356.5.

Since y=1+u+v, this is u+v<=355.5. Equality selects 352 because the secondary tie-breaker favors the candidate closer to Close.

Therefore TP2 exists exactly when u<351 and u+v<=355.5.

### C4. Combined plan feasibility

Legacy-accepted plan requires EP1, EP2, TP1, SL181, and SL361; TP2 may be absent. Exact condition:

10 < u < 360.

Full six-level plan requires EP1, EP2, TP1, TP2, SL181, and SL361. Exact conditions:

10 < u < 351 and u+v <= 355.5.

### C5. Permanently unusable EP1 ordinals

For actual adverse sign -1, EP1 ordinals 1 through 9 are permanently unusable for a complete stop pair. Ordinal 1 has no ray; 2 through 5 have neither candidate; 6 through 9 have only SL181. Every harmonic from 11 through 361 supports both stops.

Ordinal 361 is stop-capable, but an EP1 at 361 implies Close beyond it, leaving no profit-side TP1. It is not complete-plan-capable within the finite grid.

### C6. Exhaustive failure taxonomy

```
Failure | Classification | Exact condition
Invalid BUY pivot | input-dependent | Pivot >= Close
Invalid SELL pivot | input-dependent | Pivot <= Close
Non-positive Norm | input-dependent | N <= 0
Fewer than two entries | input/scale-dependent | u <= 1
EP1 has neither stop | ordinal/boundary-dependent | EP1 in 1..5
EP1 has SL181 only | ordinal/boundary-dependent | EP1 in 6..9
SL361 only | impossible in enumerated geometry | no ordinal
Complete stop pair | ordinal-dependent | harmonic EP1 >= 11
No TP1 | grid-boundary-dependent | u >= 360
TP2 absent but accepted | grid/RawTP-dependent | TP1=361
Full plan unavailable | scale/Norm/boundary-dependent | u>=351 or u+v>355.5
Negative generated price | scale/input-dependent | not rejected by source
Unsupported degree treated as 361-style | implementation defect | degrees != 181
```

## SECTION D - NORMALIZED PRICE-SPACE ANALYSIS

Define D=abs(Close-Pivot)>0, N=DailyNorm>0, and a=abs(Increment)>0. Then u=D/a and v=N/a. Close has normalized coordinate x=1+u. RawTP has normalized coordinate y=1+u+v=1+(D+N)/a.

These dimensionless values completely determine ordinal selection. Dollar prices are recovered by Pivot+(n-1)a for BUY and Pivot-(n-1)a for SELL.

The ratio D/N=u/v is determined by the inputs. It determines only the ratio between u and v, not their absolute normalized values. One continuous scale degree of freedom remains.

## SECTION E - SCALE IDENTIFIABILITY

### E1. Proof of non-identifiability

For every a>0 define P_a(n)=Pivot+s(n-1)a, where s=+1 for BUY and -1 for SELL. Every such map preserves the 19x19 ordinal spiral, harmonic membership, ray membership, ordinal ordering, and the functional form of the entry, target, and square-root stop methods.

Different a nevertheless produce different u=D/a and v=N/a and can therefore select different ordinals and prices. The transformation a -> lambda*a for any lambda>0 is an unconstrained continuous degree of freedom. No equation in the supplied algorithm establishes a=f(D,N).

Dimensional validity does not create uniqueness. Each of a=N, a=D, a=sqrt(DN), and a=D/11 has price units, but none follows uniquely from units alone. The scale is therefore not identifiable from Pivot, Close, Norm, and Direction. Infinitely many positive affine scales preserve the same underlying SQ9 geometry.

## SECTION F - FEASIBLE SCALE INTERVALS

### F1. Legacy interval

From 10 < D/a < 360 with D,a>0:

D/360 < a < D/10.

For every valid D>0 this is a nonempty continuous interval. It never selects an isolated scale. Feasibility alone cannot identify correctness or predictive value.

### F2. Full six-level interval

The conditions 10<u<351 and u+v<=355.5 become:

a < D/10,

a > D/351,

a >= (D+N)/355.5.

Thus the full scale domain is the intersection of the open interval (D/351,D/10) with the closed lower condition [(D+N)/355.5,infinity).

A full-plan scale exists exactly when (D+N)/355.5 < D/10, which reduces to N/D < 34.55. At equality the apparent scale is a=D/10, but that boundary is excluded because u=10 leaves EP1 at 9. Therefore equality is not feasible.

### F3. Possible outcomes

- Legacy feasibility: always a continuous nonempty interval for valid D; never a unique scale.
- Full-plan feasibility: a continuous interval when N/D<34.55; no feasible scale when N/D>=34.55; never a uniquely selected scale.
- Isolated feasible scales do not arise from the original inequalities.

## SECTION G - CANDIDATE COMPARISON

### Candidate 0: original magnitude a=1

Strengths: exact v0.2 output parity; simplest port; no alteration to current price semantics. Weaknesses: u=D directly, making feasibility depend on absolute dollar distance; tight low-price examples fail; wide distances reach the finite target boundary; one dollar is not part of grid geometry; the method is not cross-price invariant.

Verdict: retain only as an explicitly named Legacy $1 Unit mode.

### Candidate 1: Daily Norm unit a=N

Then v=1 and u=D/N. This is dimensionally valid and preserves geometry, but adds the semantic assumption that one ordinal step equals one Daily Norm. That assumption is not present in the source. A legacy plan would require 10<D/N<360, so Pivot-to-Close distance must exceed ten Norms. Requested tests fail for low, medium, and high examples. The wide case succeeds but produces extremely distant dollar stops.

Verdict: do not use as default. The absence of a new numerical constant does not make the semantic assumption uniquely justified.

### Candidate 2: explicit close-coordinate calibration

Choose a fixed normalized displacement u0 and set a=D/u0, imposing x=1+u0.

Recommended deterministic convention when no extra field is allowed: u0=11, hence a=D/11. This places Close at x=12 and gives EP1=11, EP2=9, SL181=6, and SL361=2.

Why 11 is the narrowest operational choice: EP1 11 is the first harmonic supporting both stops; strict eligibility requires x>11; x=12 is the next integer coordinate; therefore u0=11. It is the coarsest scale in this simple integer-coordinate class that supplies both stops.

Trade-offs: it standardizes entry geometry across stocks; does not establish predictive meaning; can produce large dollar stops when D is wide; may produce non-positive levels; and changes output parity. It is an operational calibration, not a mathematical discovery.

### Market-defined quotation increment

A valid quotation increment is relevant to final validation and display but generally too small to solve finite grid coverage. At a one-cent scale, D=$5 produces u=500 and no target. A universal one-cent assumption is also inappropriate across every US-listed security and venue context.

Verdict: use a known instrument increment for validation and presentation, not automatically as the SQ9 affine scale.

## SECTION H - EXECUTED NUMERICAL TESTS

SELL equivalents retain the requested Close and Norm and reflect the same D above Close.

```
Case | BUY Pivot | Close | SELL Pivot | Norm
Low | 24.50 | 24.80 | 25.10 | 0.60
Medium | 175.00 | 180.00 | 185.00 | 4.00
High | 950.00 | 970.00 | 990.00 | 15.00
Wide | 40.00 | 390.00 | 740.00 | 20.00
```

Distances below are absolute dollar distances from EP1.

### H1. Original fixed scale a=1

Low BUY and SELL fail because u=0.3 supplies fewer than two entries.

Medium BUY and SELL reach EP1=5 and fail both stops.

High BUY and SELL succeed fully.

Wide BUY and SELL are accepted legacy plans with TP2 unavailable.

High BUY:

Scale 1; Start 950; Increment +1; EP1 n19 $968; EP2 n17 $966; RawTP $985; TP1 n37 $986; TP2 n40 $989; SL181 n11 $960; SL361 n6 $955; EP1-to-TP1 $18; EP1-to-SL181 $8; EP1-to-SL361 $13.

High SELL:

Scale 1; Start 990; Increment -1; EP1 n19 $972; EP2 n17 $974; RawTP $955; TP1 n37 $954; TP2 n40 $951; SL181 n11 $980; SL361 n6 $985; distances $18, $8, and $13.

Wide BUY:

Scale 1; Start 40; Increment +1; EP1 n343 $382; EP2 n334 $373; RawTP $410; TP1 n361 $400; TP2 unavailable; SL181 n307 $346; SL361 n273 $312; distances $18, $36, and $70. calculate_plan accepts this plan.

Wide SELL:

Scale 1; Start 740; Increment -1; EP1 n343 $398; EP2 n334 $407; RawTP $370; TP1 n361 $380; TP2 unavailable; SL181 n307 $434; SL361 n273 $468; distances $18, $36, and $70. calculate_plan accepts this plan.

### H2. Daily Norm scale a=N

Low BUY and SELL fail entry selection. Medium and High BUY and SELL select EP1=2 and fail stops. Only Wide succeeds.

Wide BUY:

Scale 20; Start 40; Increment +20; EP1 n17 $360; EP2 n15 $320; RawTP $410; TP1 n19 $400; TP2 n21 $440; SL181 n9 $200; SL361 n5 $120; distances $40, $160, and $240.

Wide SELL:

Scale 20; Start 740; Increment -20; EP1 n17 $420; EP2 n15 $460; RawTP $370; TP1 n19 $380; TP2 n21 $340; SL181 n9 $580; SL361 n5 $660; distances $40, $160, and $240.

### H3. Explicit calibration a=D/11

Low BUY:

Scale 0.0272727273; Start 24.50; Increment +0.0272727273; EP1 n11 $24.77272727; EP2 n9 $24.71818182; RawTP $25.40; TP1 n34 $25.40; TP2 n37 $25.48181818; SL181 n6 $24.63636364; SL361 n2 $24.52727273; distances $0.62727273, $0.13636364, and $0.24545455.

Low SELL:

Scale 0.0272727273; Start 25.10; Increment -0.0272727273; EP1 n11 $24.82727273; EP2 n9 $24.88181818; RawTP $24.20; TP1 n34 $24.20; TP2 n37 $24.11818182; SL181 n6 $24.96363636; SL361 n2 $25.07272727; same distances as BUY.

Medium BUY:

Scale 0.4545454545; Start 175; Increment +0.4545454545; EP1 n11 $179.54545455; EP2 n9 $178.63636364; RawTP $184; TP1 n21 $184.09090909; TP2 n23 $185; SL181 n6 $177.27272727; SL361 n2 $175.45454545; distances $4.54545455, $2.27272727, and $4.09090909.

Medium SELL:

Scale 0.4545454545; Start 185; Increment -0.4545454545; EP1 n11 $180.45454545; EP2 n9 $181.36363636; RawTP $176; TP1 n21 $175.90909091; TP2 n23 $175; SL181 n6 $182.72727273; SL361 n2 $184.54545455; same distances as BUY.

High BUY:

Scale 1.8181818182; Start 950; Increment +1.8181818182; EP1 n11 $968.18181818; EP2 n9 $964.54545455; RawTP $985; TP1 n21 $986.36363636; TP2 n23 $990; SL181 n6 $959.09090909; SL361 n2 $951.81818182; distances $18.18181818, $9.09090909, and $16.36363636.

High SELL:

Scale 1.8181818182; Start 990; Increment -1.8181818182; EP1 n11 $971.81818182; EP2 n9 $975.45454545; RawTP $955; TP1 n21 $953.63636364; TP2 n23 $950; SL181 n6 $980.90909091; SL361 n2 $988.18181818; same distances as BUY.

Wide BUY:

Scale 31.8181818182; Start 40; Increment +31.8181818182; EP1 n11 $358.18181818; EP2 n9 $294.54545455; RawTP $410; TP1 n13 $421.81818182; TP2 n15 $485.45454545; SL181 n6 $199.09090909; SL361 n2 $71.81818182; distances $63.63636364, $159.09090909, and $286.36363636.

Wide SELL:

Scale 31.8181818182; Start 740; Increment -31.8181818182; EP1 n11 $421.81818182; EP2 n9 $485.45454545; RawTP $370; TP1 n13 $358.18181818; TP2 n15 $294.54545455; SL181 n6 $580.90909091; SL361 n2 $708.18181818; same distances as BUY.

The wide case demonstrates that normalized feasibility does not imply practically attractive dollar stops.

# SECTION I - US STOCKS RECOMMENDATION

### I1. Operational decision path

Preferred design if one additional manual field is acceptable:

1. Add a positive manual SQ9 Scale input a.
2. Validate Pivot, Close, Direction, and Norm exactly as v0.2.
3. Assign BUY Increment +a and SELL Increment -a.
4. Calculate u=D/a and v=N/a.
5. Disclose whether it satisfies legacy 10<u<360 and full-plan 10<u<351 with u+v<=355.5.
6. Execute the original EP/TP/SL algorithms without fallback.

7. Display the supplied scale and its calibration basis with the plan.

This is the most transparent solution because it exposes, rather than conceals, the missing scale degree of freedom.

If the specified input form cannot add a scale field, use a=D/11 and label the mode: Normalized close-coordinate calibration; Close x=12; EP1 anchored to first complete-stop harmonic n=11.

This recommendation is not uniquely true, predictive, or historically optimized. It is deterministic, affine, direction-symmetric, independent of absolute price, compatible with the existing ordinal rules, and based on the first harmonic carrying both mandated angular stops.

Retain a=1 as a separately named Legacy $1 Unit mode for reproducibility and output-parity checks.

### I2. Algorithm preservation versus output parity

Changing a preserves the canonical spiral, ordinal coordinates, harmonic membership, rays, entry ordering, target minimization, RawTP construction, angular stop method, and failure logic. It does not preserve Close coordinate, selected EP/TP ordinals, dollar outputs, or success/failure for a specific input. This is algorithm preservation without output parity.

### I3. Practical safeguards

The calibration can generate stops below zero for BUY or implausibly high SELL stops when Pivot-to-Close distance is wide. The source does not reject them. A future calculator should identify non-positive equity prices as invalid market outputs. That is domain validation, not a new price-level formula.

### I4. Ten-day holding period

The ten-trading-day limit has no mathematical role in the supplied price mapping. The engine contains no time coordinate, bar count, or date-dependent scale. It must remain separate as plan metadata or a manual expiry instruction. No time-based SQ9 scaling can be derived from this source.

## SECTION J - MINIMUM PHP PORT REQUIREMENTS

No PHP implementation is supplied. The minimum mathematical specification is:

1. Reproduce the exact 19x19 orientation and ordinal map.
2. Preserve all 73 harmonics.
3. Preserve strict less-than and greater-than comparisons.
4. Preserve TP1 tuple ordering: RawTP distance, then Close distance.
5. Preserve nullable TP2 and the legacy accepted behavior.
6. Restrict internal angular calls to exactly 181 or 361.
7. Preserve 181 as opposite ray/root delta 1 and 361 as same ray/root delta 2.
8. Preserve nearest available ray-candidate selection.
9. Do not round before ordinal selection or tie-breaking.
10. Separate internal calculation precision, quotation-increment validation, and display formatting.

For financial precision, parse user prices as decimal strings and retain sufficient internal precision. Compute roots at higher working precision. Apply deterministic tie behavior equivalent to Python tuple ordering. Round only final display values. Validate displayed or executable prices against a known instrument quotation increment when supplied. Do not assume a universal one-cent increment.

The Python close_n tolerance of 1e-8 affects only reported close_n, not EP/TP/SL selection. A PHP port must not let display rounding change strict eligibility.

## SECTION K - FINAL THREE-GATE VERDICT

### GATE 1: Was the actual Python v0.2 calculation engine audited?

PASS.

Evidence: exact spiral construction was executed; all 361 cells and ordinals were verified; all 73 harmonics were enumerated; strict entries, targets, tie-breaking, ray selection, adverse sign, angular stops, and nullable TP2 behavior were analyzed directly. Unsupported-degree behavior was identified from the supplied conditional.

### GATE 2: Were EP/TP/SL mathematical limitations comprehensively mapped?

PASS.

Evidence: every harmonic was classified through exact ring formulas; both adverse signs were exhaustively evaluated; stop counts are 64 both, 4 SL181 only, 0 SL361 only, and 5 neither; strict entry, target, stop-pair, legacy, and full-plan regions were derived; BUY/SELL symmetry was proven; requested numerical cases and alternative mappings were executed.

### GATE 3: Was the US equities price mapping resolved to a mathematically justified conclusion and acti

PASS.

Evidence: non-identifiability was proven as a continuous affine degree of freedom; exact feasible scale intervals were derived; original, Norm-unit, quotation-tick, and affine-calibration candidates were compared; the unavoidable external assumption was identified; a deterministic a=D/11 default and a preferred user-supplied scale path were provided without predictive claims.

## FINAL DECISION TABLE

| QUESTION | ANSWER | EVIDENCE |
|---|---|---|
| 1. Is increment +/-1 part of SQ9 geometry? | No. Sign supplies price direction and magnitude 1 supplies an affine price unit. | Spiral and harmonics are constructed before price mapping. |
| 2. Does changing increment preserve the original 19x19 harmonic structure? | Yes, under a single nonzero affine increment. It can change selected ordinals and outputs. | Harmonics depend only on coordinates; levels are assigned afterward. |
| 3. Which EP1 ordinals support both angular stops? | All 64 harmonic ordinals from 11 through 361 under validated adverse sign -1. | Exhaustive enumeration; 1-5 support neither and 6-9 only SL181. |
| 4. What is the minimum normalized displacement required for a legacy-accepted trading plan? | Strictly u>10, plus u<360. Ten is an infimum, not an attained minimum. | At u=10 strict entry comparison excludes ordinal 11 and leaves EP1=9 without SL361. |
| 5. Does feasibility differ between BUY and SELL? | No under validated pivot relationships. | Both yield x=1+D/a, target n>x, and adverse sign -1. |
| 6. Can a unique equity increment be derived from existing inputs alone? | No. | Any a>0 is an affine embedding; D/N fixes only u/v. |
| 7. Can feasible increment intervals be calculated? | Yes. Legacy D/360<a<D/10. Full: a>D/351, a>=(D+N)/355.5, and a<D/10. | Direct algebra from normalized feasibility. |
| 8. What explicit assumption is required to select one increment? | A scale-calibration condition, either supplied manually or fixed as policy. | Feasibility produces an interval, not one point. |
| 9. What is the recommended equity price mapping? | Prefer a documented user-supplied a. Without a new field, use a=D/11; retain a=1 as Legacy mode. | D/11 makes x=12 and EP1=11, the first complete-stop harmonic. |
| 10. What is the minimum mathematical modification for PHP/HTML? | Parameterize magnitude instead of forcing 1 while preserving all ordinal selection rules; add decimal-safe calculation, degree validation, nullable TP2, and scale disclosure. | Scale is the only unresolved mathematical degree of freedom. |

# RESEARCH DISCIPLINE AND LIMITATIONS

This assessment separates four concepts:

1. Mathematical validity: the grid, mappings, inequalities, and candidate rules are internally defined and auditable.
2. Algorithmic feasibility: exact normalized domains show when the code can produce its required fields.
3. Practical usability: generated prices can still be too distant, non-positive, or inconsistent with an instrument's quotation increment.
4. Empirical edge: no conclusion about profitability, hit rate, or predictive value follows from mathematical feasibility.

No scale was selected by maximizing apparent profit, risk/reward, or historical performance. No grid expansion, fallback target, alternative indicator, machine-learning model, volatility rule, or trading rule has been introduced.

# VERIFICATION RECORD

The following checks were executed in temporary standalone Python sessions:

PASS: exact 19x19 build; 361 unique coordinates and ordinals; range 1..361.

PASS: documented 3/4/5, 2/1/6, 9/8/7 orientation.

PASS: 73 harmonics; 37 cardinal-inclusive-center; 37 diagonal-inclusive-center; intersection center only.

PASS: nine harmonics per noncentral ray.

PASS: harmonic gap distribution of eight each for gaps 1 through 9.

PASS: exhaustive SL181 and SL361 evaluation for every harmonic and both adverse directions.

PASS: requested low, medium, high, and wide BUY and SELL examples at a=1.

PASS: the same examples at a=Norm and a=D/11.

PASS: algebraic interval results cross-checked against the exact source behavior.

End of report.